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Perceptrons and Margins

- Number of mistakes depends on the best possible w^*'s margin.
- w_{perc} need not necessarily be w^*. It could be the one with smaller margin also.

Goal: To come up with a formulation that maximizes 'margin'

$$\max_{w,\gamma} \quad \gamma$$

such that, $(w^T x_i) \cdot y_i \geq \gamma \quad \forall i$

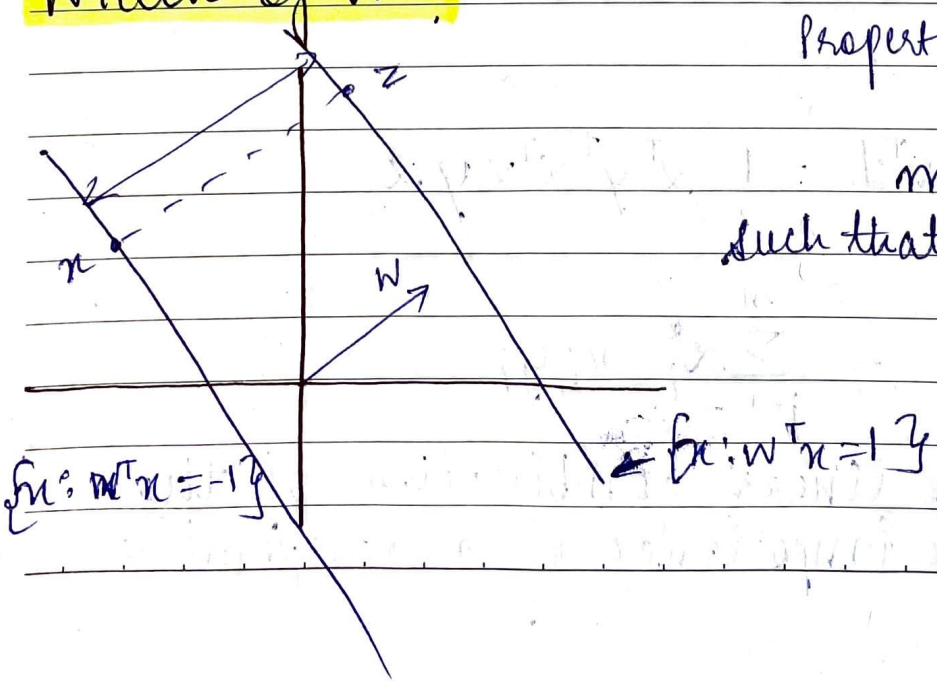
$$\|w\|^2 = 1$$

Modified equivalent formulation:

$$\max_{w} \quad \text{width}(w)$$

such that, $(w^T x_i) y_i \geq 1 \quad \forall i$

Width of w ?



Property $\Rightarrow$ $w^T x = -1$

$$\min_{z} \|x - z\|^2$$

such that, $w^T z = +1$

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Solution

$$\text{width}(w) = \frac{2}{\|w\|^2}$$

Final formulation - Primal

$$\min_{w} \frac{1}{2}\|w\|^2$$

such that, $(w^T x_i) y_i \geq 1 \quad \forall i$

$$\Rightarrow \underbrace{1-(w^T x_i) y_i}_{g_i(w)} \leq 0$$

$$L(w, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i \left(1 - (w^T x_i) y_i\right)$$

$$\min_{w} \max_{\alpha \geq 0} \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i \left(1 - (w^T x_i) y_i\right)$$

$$=$$

$$\max_{\alpha \geq 0} \min_{w} \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i \left(1 - (w^T x_i) y_i\right)$$

Dual Problem

$$\max_{\alpha \geq 0} \alpha^T 1 - \frac{1}{2} \alpha^T y^T x^T x y \alpha$$

$$w^*_{\alpha^*} = \sum_{i=1}^{n} \alpha_i^* x_i y_i$$

$\Rightarrow$ optimal w^* is a linear combination of the data points where importance of a datapoint is given by α_i^*